

Faculty Overview

Benefits / limitations

A reference for IT faculty planning coursework

● INCLUDED

○

○

WHAT'S INCLUDED

IaaS

- Azure VMs (Windows / Linux), VNets, NSGs
- Public IPs (Standard SKU), managed disks
- Container Instances; AKS at small node counts
- Storage: Blob, Files, Queues, Tables

PaaS

- App Service, Container Apps, Functions
- Azure SQL serverless, PostgreSQL Flex, Cosmos DB
- Service Bus, Event Grid, Event Hubs, API Mgmt
- Azure AI Services, OpenAI, ML, Foundry

SaaS / subscriptions

- Visual Studio 2022 Enterprise IDE
- M365 E5 Developer Tenant (25 seats)
- Microsoft Foundry (model catalog, serverless)
- Microsoft Learn training credit
- Windows 10 / 11 Enterprise dev / test keys
- Windows Server + SQL Server dev / test keys
- GitHub Enterprise (verify entitlement)

Self-administration

- Global Admin on the sandbox tenant
- Invite students as guests; scope to a single RG
- Manage IAM, budgets, tags, IaC without a ticket
- Cleanup automation in scripts / powershell

Pinned region

- Canada East for all labs
- Consistent SKU availability across cohorts
- Removes the trial-region randomization problem

WHAT'S NOT INCLUDED

Governance gaps

- Outside NSCC central IT (no Intune / DLP / SIEM)
- No federation with NSCC SSO; guest accounts only
- No backup or retention beyond per-resource config
- Faculty own incident response and audit
- No transferred FOIPOP / PHIA / PCI posture

Missing licenses

- Defender for Cloud paid plans (Servers, SQL, etc.)
- Microsoft Sentinel (no SIEM / SOAR)
- Defender for Endpoint beyond the 25-seat tenant
- Microsoft 365 Copilot / GitHub Copilot Business
- GitHub Advanced Security
- Entra ID Governance (beyond included P2)
- Teams Premium
- Microsoft Purview Data Governance (Azure-side)
- ExpressRoute, dedicated hosts, reserved instances
- Most paid Marketplace offerings

PAID / EATS CREDIT FAST

- Azure Firewall (~CA\$1.40 / hour base)
- Azure Bastion (~CA\$0.19 / hour / host)
- Azure DDoS Protection Standard (~CA\$4k / month)
- Front Door Premium, Private Endpoints at scale

Hard ceilings

- ~CA\$200 / mo credit per subscriber
- M365 E5 dev tenant capped at 25 users
- Dev tenant auto-renews with activity, not unconditionally
- No production traffic, no real student PII

Administrative burden

Faculty are simultaneously tenant admin, billing watcher, IAM operator, incident responder, and cleanup enforcer. No central handoff path exists inside NSCC IT for issues that arise inside the sandbox.